

PROG1327B - R II: Data Analysis and Visualization

Assignment I (Group Assignment)

Data Preparation for Machine Learning

Objective:

To prepare a dataset for a machine learning project, which will require the prediction of influence scores of scientific journals (regression problem) in Assignment II. At this stage, the assignment will only focus on the process of data cleaning.

Instructions:

This assignment is worth 20 points in total. To get 12 points, you must complete the required tasks listed below:

1. Identify numeric and categorical variables and their data types. Identify the number of columns and rows per file. **(1 point)**
2. Identify the column to join the data files and the join type. Perform the operations to compile a single dataset. Identify the resulting number of rows and columns. **(2 points)**
3. Identify the number/percent of missing data per column. Decide whether any columns should be dropped based on these results and drop them if necessary. Additionally, drop any columns that you consider irrelevant for the project. **(2 points)**
4. Identify any data inconsistencies and fix them (e.g.: datatype conversion, string reformatting, invalid data, etc.). **(2 points)**
5. Based on point 3, perform data inputting on the columns that require it. Optionally, you can remove rows with missing values. **(2 points)**
6. Detect and remove duplicates if present. Identify potential outliers in your dataset and decide whether to keep or remove them. **(2 points)**
7. Give a summary of your final dataset, including the number of rows and columns as well as the variables and datatypes. **(1 point)**

The remaining 8 points will be awarded for the quality of your explanations, including the rationale and justification for each decision you make during the process **(1.5 points per task, with exception of step 1)**.

Deliverables:

- An RMD file containing the code and preliminary conclusions of your exploratory data analysis.
- The corresponding pdf or html file generated from the above RMD file.
- The cleaned dataset in csv format.

Note: *ALL files need to be submitted to receive a mark. If a file is missing or does not compile (in case of the RMD file), the corresponding delay penalty will be applied (20% of the maximum value per day late).*

Due Date:

- October 7, 11:59 pm.

Teams:

- Groups of two.

Dataset Description:

The dataset contains information about scientific journals and publishers. The data comes from different sources, including researchers, web scraping and the publishers themselves. It comprises three different files: *api_journal11-13-17.csv*, *api_price11-13-17.csv* and *estimated-article-influence-scores-2015.csv*. Some variables the ISSN, Journal title, subscription price, number of citations and number of published papers, among others. The output variable is the projected article influence, which can be interpreted as the expected influence of an article published by a given journal.

Data Files:**File 1:** *api_journal11-13-17.csv*

1. *Issn*: The International Standard Serial Number of the publication.
2. *Journal-name*: The name of the scientific journal.
3. *Pub_name*: The name of the publisher.
4. *Is_hybrid*: Electronic and printed versions of journal (1); only electronic version of journal (0).

5. *category*: The category or scientific field of the journal.
6. *url*: The web page address of the journal.

File 2: *api_price11-13-17.csv*

1. *id*: Observation id.
2. *price*: The subscription's price.
3. *date_stamp*: The date in which the information was collected.
4. *Journal_id*: The International Standard Serial Number of the publication.
5. *Influence_id*: The influence Id.
6. *url*: The web page address of the journal.
7. *license*: Rights for publication, distribution, and use of research.

File 3: *estimated-article-influence-scores-201.csv*

1. *Journal_name*: The name of the scientific journal.
2. *issn*: The International Standard Serial Number of the publication.
3. *Citation_count_sum*: The total number of citations of journal.
4. *Paper_count_sum*: The total number of papers published by the journal.
5. *Avg_cites_per_paper*: The average number of citations per paper.
6. *Proj_ai*: The projected article influence. The higher the influence, the better the scientific credibility of the journal.
7. *Proj_ai_year*: The year of projected article influence